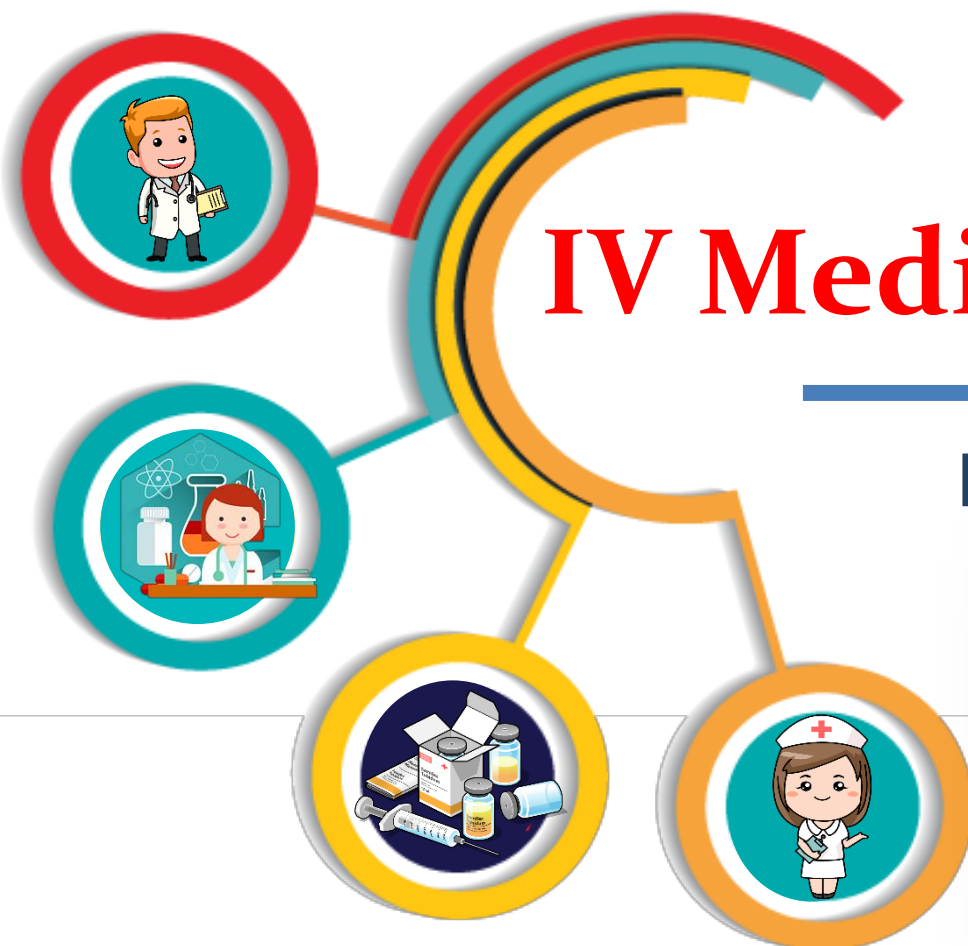
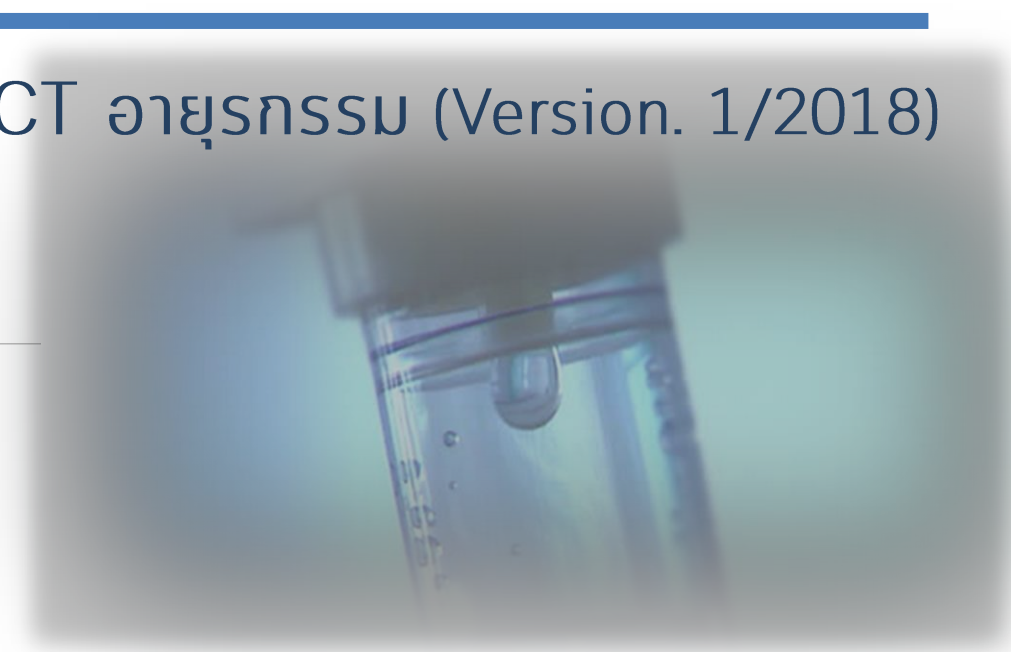




IV Medication Drip Chart

PCT อายุรกรรม (Version. 1/2018)





Index

Summary chart	2	Lidocaine	3, 9
IV Medication drip chart		Midazolam	3, 9
Amiodarone	2	Milrinone	3, 10
Atracurium	2, 5	Morphine	3, 10
Cisatracurium	2, 5	Nicardipine	4, 10
Dexmedetomidine	2	Nitroglycerine	4, 11
Dobutamine	2, 6	Nitroprusside	4, 11
Dopamine	2, 6	Norepinephrine	4, 12
Epinephrine	2, 7	Pancuronium	4
Esmolol	2, 7	Propofol	4, 12
Etomidate	3	Rocuronium	4, 13
Fentanyl	3, 8	Succinylcholine	4
Heparin	3, 8	Vecuronium	4, 13
Insulin	3	IV compatibility chart	14
Labetalol	3		



Drug name	Administration	Dosing instruction	How to titrate	Monitoring/Other information
Amiodarone (Cordarone® 150mg/3ml)	150mg/100mL 600-900 mg/500mL Max conc: 2mg/mL Solution: D5W only	ACLS : 300 mg bolus then 150 mg over 10 min Infusion: 1 mg/min x 6 h then 0.5 mg/min x18 h Max dose 2.2 g/day Max rate 30 mg/min	Non-titratable Peripheral line : Up to 2 mg/mL Central line : Up to 6 mg/mL	BP (hypotension occurs frequently with initial rates), HR (arrhythmias: AV block, bradycardia, VT/VF, torsades de pointes), Electrolytes, Pulmonary function test, Thyroid function , Liver enzymes , Venous irritation (>2 mg/mL)
Atracurium (Tracrium® 50 mg/5ml)	250mg/100ml (2.5mg/ml) Max conc : Undiluted Solution: NSS, D5W, D5NS	Bolus: 0.4-0.5 mg/kg Infusion: 5-20µg/kg/min (see drip chart)	Titrate dose by 1-2 µg/kg/min to maintain desired neuromuscular blockade	BP, HR, peripheral nerve stimulation, respirations Opioid analgesic : Possibly additive histamine release and increased risk and severity of bradycardia and hypotension
Cisatracurium (Nimbex® 10mg/5mL)	100mg/100mL (1mg/ml) Max conc : Undiluted Solution: NSS, D5W, D5NS	Bolus: 0.1-0.2 mg/kg Infusion: 1-10 µg/kg/min (see drip chart)	Titrate dose by 2 µg/kg/ min q 10 mins to maintain desired neuromuscular blockade	BP, HR, peripheral nerve stimulation, muscle twitching, blood gases, respirations
Dexmedetomidine (Precedex® 200µg/2ml)	400µg/100mL Max conc. 4µg/mL Solution: NSS, D5W	Load: 1 µg/kg over 10 min Infusion: 0.2 – 1.5 µg/kg/h	Titrate dose by 0.1 µg/kg/h q 15 mins to achieve a RASS score of 0 to -1. Titrate slowly to avoid hypotension.	Do not bolus this drug due to bradycardia, hypotension
Dobutamine (250mg/20mL)	500mg/500mL (1mg/ml) Max conc: 5mg/mL Solution: NSS, D5W, D5NS, D5S/2	Infusion: 2.5-15 µg/kg/min (up to 20 µg/kg/min if on beta blockers) (see drip chart)	Titrate dose by 2.5-5 µg/kg/min q 5-15 mins to achieve a MAP ≥65mmHg.	BP, HR, EKG, Urine output Infuse via central line to avoid extravasation
Dopamine (500 mg/10ml)	500mg/500mL (1mg/ml) 1000mg/500ml (2 mg/ml) Max conc: 3.2mg/mL Solution: NSS, D5W, D10W, D5NS, D5NS/2	Infusion: 2-20 µg/kg/min Cardiac dose 5-10 µg/kg/min Vasopressor 10-20 µg/kg/min (see drip chart)	Titrate dose by 2-5 µg/kg/min q 5-15mins to achieve a MAP ≥65 mmHg.	BP, HR, EKG, Urine output, Sign of Peripheral necrosis Infuse via central line to avoid extravasation
Epinephrine (Adrenalin® 1mg/ml)	10mg/100mL (0.1mg/ml) Max conc : Undiluted Solution: NSS, D5W, D5S	ACLS : 1 mg IV q 3-5 min Vasopressor Infusion : 1-10 µg/min (0.01-3 µg/kg/min) (see drip chart)	Titrate dose by 0.01 µg/kg/min q 15 mins to achieve a MAP of ≥ 65 mmHg	BP, HR, Urine output, Sign of Peripheral necrosis Tachyarrhythmias and hypertension at high doses
Esmolol (Esbloc® 100mg/10ml)	1000mg/100mL (10mg/ml) Max conc : Undiluted Solution: NSS, NS/2, D5W, D5S, D5S/2, D5NS	Bolus: 0.5-1mg/kg in 1 min Infusion: 0.05-0.3 mg/kg/min (see drip chart)	Titrate dose by 0.05 mg/kg/min q 4 mins to desired effect Should taper off slowly.	BP, HR , EKG, Phlebitis, Inflammatory skin Contraindication : Sinus bradycardia, Heart block>1 st degree, Cardiogenic shock, Decompensated HF Precaution : Hypotension, Bronchospasm, Renal impairment Infuse via central line to prevent extravasation



IV Medication drip chart. Version 1/2018.

Drug name	Administration	Dosing instruction	How to titrate	Monitoring/Other information
Etomidate (20 mg/10ml)	20mg/10mL Max conc : Undiluted	Bolus: 0.3 mg/kg IV push over 15-60 secs	For rapid sequence intubation	BP, Myoclonic, Thrombophlebitis
Fentanyl (100µg/2mL)	200µg/100mL (2 µg/ml) Max conc : Undiluted Solution: NSS, D5W	Bolus: 0.35-0.5 µg/kg Infusion rate: 0.5-10 µg/kg/h (25– 500 µg/h) (see drip chart)	Titrate dose by 25 µg/h q 30 mins to achieve desired effect	BP, HR, RR, Pain score, Sedative score, Anaphylactoid, GI side effect
Heparin (25000u/5ml)	25,000units (5 ml)/250mL 10,000units (2 ml)/100mL Solution: NSS (24h), D5W (8h)	Weight based or aPTT based (see heparin protocol)	Based upon aPTT and per protocol	CBC, Coagulogram, Sign of bleeding
Insulin (Regular 1000unit/vial)	100units/100mL (1unit/ml) Solution: NSS		Based upon blood glucose level	Blood glucose, Potassium level (Do not lower Blood glucose > 100 mg/dL/hr. Rapid lowering may lead to cerebral edema)
Labetalol (25mg/5ml)	250mg/250ml (1mg/ml) Max conc : Undiluted Solution: NSS, D5W	Bolus: 10-20 mg (0.2mg/kg/dose) initially over 2 min then 40-80 mg q 10-20 min up to 300 mg Infusion: 0.5-3mg/min	Titrate dose q 15 mins to desired effect	BP, HR Contraindication : Sinus bradycardia, Heart block>1st degree, Cardiogenic shock, Decompensated HF Don't exceed total dose of 300 mg; duration of action extend to nearly 18 h
Lidocaine (40mg/2mL)	1000mg/250ml 2000mg/250mL Max conc : Undiluted Solution: NSS, D5W	Bolus : 1-1.5 mg/kg (2-3 mins) ; repeat dose 0.5-0.75 mg/kg in 3-5 min (Max cumulative dose: 3 mg/kg) Infusion : 1-4 mg/min (see drip chart)	Increase by 1 mg/min up to 4 mg/min until arrhythmia terminated: stop infusion once rhythm stabilizes	EKG, HR, BP, CNS effect (confusion, nervousness, seizure), Cardiovascular collapse, arrhythmia. Decrease dose by 50% for CHF, impair liver function, elderly, shock
Midazolam (Dormicum®5mg/mL)	20 mg/100ml (0.2 mg/ml) 50 mg/100mL (0.5 mg/ml) Max conc : Undiluted Solution: NSS, D5W	Bolus: 0.03-0.3 mg/kg (0.5-4mg) Infusion: 0.02-0.2 mg/kg/h (1-30 mg/h) (see drip chart)	Titrate dose by 1 mg/h every 1 hour to achieve desired effect	Drowsiness, impaired memory or coordination, agitation (paradoxical), Extravasation
Milrinone (Primacor® 10mg/10ml)	20mg/100mL (0.2mg/ml) Max conc: 0.2mg/mL Solution: NSS, D5W, NS/2	Bolus: 50 µg/kg over 10 min Infusion: 0.375 – 1 µg/kg/min (1.13 mg/kg/day) (see drip chart)	Dose increases per CO/CI or SVO ₂ and can be discontinued without weaning	BP, HR, EKG, Urine output
Morphine (10mg/ml)	100mg/100mL(1mg/mL) Max conc: 5mg/mL Solution: NSS, D5W, D10W	Bolus: 2-5 mg (0.05-0.2mg/kg) Infusion: 0.8-80mg/h (0.01-0.03 mg/kg/hr) (see drip chart)	Titrate dose by 1 mg/h q 30 min to achieve desired effect	BP, RR, HR, GI effects Pain scale assessment for analgesia. Recommend lower initial dose for renal or hepatic dysfunction and elderly



Drug name	Administration	Dosing instruction	How to titrate	Monitoring/Other information
Nicardipine (2mg/2ml)	10 mg/100 mL Max conc: 0.2mg/mL Solution: NSS, D5W, D5NS	Infusion: 5 - 15 mg/h (see drip chart)	Titrate dose by 2.5 mg/h q 15 mins to achieve BP. Reduce infusion to 3 mg/hr after BP is controlled.	BP, HR, N/V, Headache Rotate the IV infusion site q 12h if administered via a peripheral IV and q 24 hr via central line.
Nitroglycerine (50mg/10ml)	50mg/250mL (0.2 mg/mL) Max conc: 0.4mg/mL Solution: NSS, D5W	Infusion: 5 - 400 µg/min (0.25-1 µg/kg/min) (see drip chart)	Titrate dose by 5 µg/min q 5 mins to achieve desired effect.	BP, HR, Headache, Flushing, Postural hypotension, Reflex tachycardia, dizziness, Methemoglobinemia Tolerance may develop after 12-24 h
Nitroprusside (50mg/vial)	50mg/250mL (0.2mg/mL) Max conc: 1 mg/mL Solution: D5W only	Infusion: 0.5 – 10 µg/kg/min **Protect from light** (see drip chart)	Titrate dose by 0.25 µg/kg/min q 3 mins to achieve BP.	BP, HR, Flushing, Methemoglobinemia, Cyanide poisoning Impaired renal function or doses > 3mcg/kg/min for > 72 hours may cause thiocyanate toxicity
Norepinephrine (4mg/4ml)	4mg/100mL Max conc: 1.28mg/mL Solution: D5W only	Infusion : 0.02-3 µg/kg/min or 0.5-30 µg/min (see drip chart)	Titrate dose by 0.05 µg/kg/min q 3 mins to achieve a MAP of ≥65 mmHg.	EKG, HR, BP, Peripheral ischemia, Urine output Infuse via central line to avoid extravasation
Pancuronium (Pavulon® 4mg/2ml)	Max conc : Undiluted Solution: NSS, D5W, D5NS	Bolus: 0.05-0.1 mg/kg then 0.015-0.1mg/kg q 30-60 min Infusion: 0.8-1.7 µg/kg/min (0.06-0.1mg/kg/hr)		BP, HR, RR **Prolong blockade
Propofol (Diprivan® 200mg/20ml)	1000mg/100mL Max conc : Undiluted Min conc: 2mg/ml Solution: D5W	Bolus : 1-3mg/kg Infusion : 5-100 µg/kg/min (see drip chart)	Titrate dose by 5 µg/kg/min q 5 mins to achieve RAAS score 0 to -2 if use with paralytics, titrate to a bispectral index (BIS) of 40 to 60	BP, HR Propofol infusion syndrome (metabolic acidosis (base deficit > 10 mmol/L), rhabdomyolysis, hyperlipidemia, and enlarged or fatty liver)
Rocuronium (Esmeron® 50mg/5ml)	100mg/100mL Max conc : Undiluted Solution: NSS, D5W, D5S	Bolus: 0.6-1.2mg/kg Infusion: 0.4-0.7 mg/kg/h (see drip chart)	Titrate dose to achieve desired effect.	BP, HR, RR, QT prolong
Succinylcholine (500mg/vial)	Conc: 1-2mg/mL Solution: NSS, D5W, D5NS	Bolus: 0.3-1.1 mg/kg then 0.04 - 0.07 mg/kg q 5-10 min Infusion: 2.5 mg/min Max dose: 150 mg	Continuous IV infusion for long surgical procedures	BP, HR, Malignant hyperthermia electrocardiogram neuromuscular function
Vecuronium (Vecuron® 10 mg/vial)	100mg/250mL (0.4mg/ml) Max conc: 2mg/ml Solution: NSS, D5W, D5NS	Bolus: 0.05-0.1 mg/kg Initial: 0.8-2 µg/kg/min (see drip chart)	Titrate dose by 0.2 µg/kg/min q 10 mins to achieve desired effect.	BP, HR, RR **Prolong blockade



Atracurium (Tracrium® 50 mg/5ml)

Max concentration : Undiluted (10mg/ml)

Solution : NSS, D5W, D5NS

Dosing instruction : Bolus : 0.4 - 0.5 mg/kg

Infusion Rate : 5 – 20 µg/kg/min

Titrate dose by 1-2 µg/kg/min to maintain desired neuromuscular blockade

Infusion chart : Atracurium 250mg/100mL (2.5 mg/mL)

Dose (µg/kg/min)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
5	5	5	6	7	7	8	8	9	10
6	6	6	7	8	9	9	10	11	12
7	7	8	8	9	10	11	12	13	13
8	8	9	10	11	12	12	13	14	15
9	9	10	11	12	13	14	15	16	17
10	10	11	12	13	14	16	17	18	19
11	11	12	13	15	16	17	18	20	21
12	12	13	14	16	17	19	20	22	23
13	12	14	16	17	19	20	22	23	25
14	13	15	17	18	20	22	24	25	27
15	14	16	18	20	22	23	25	27	29
16	15	17	19	21	23	25	27	29	31
17	16	18	20	22	24	27	29	31	33
18	17	19	22	24	26	28	30	32	35
19	18	21	23	25	27	30	32	34	36
20	19	22	24	26	29	31	34	36	38

Cisatracurium (Nimbex® 10mg/5ml)

Max concentration : Undiluted (2mg/ml)

Solution : NSS, D5W, D5NS

Dosing instruction : Bolus Dose : 0.1-0.2mg/kg

Infusion Rate : 1 – 10 µg/kg/min

Titrate dose by 2 µg/kg/min q10mins to maintain desired neuromuscular blockade

Infusion chart : Cisatracurium 100mg/100mL (1 mg/mL)

Dose (µg/kg/min)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
1	2	3	3	3	4	4	4	5	5
1.5	4	4	5	5	5	6	6	7	7
2	5	5	6	7	7	8	8	9	10
2.5	6	7	8	8	9	10	11	11	12
3	7	8	9	10	11	12	13	14	14
4	10	11	12	13	14	16	17	18	19
5	12	14	15	17	18	20	21	23	24
6	14	16	18	20	22	23	25	27	29
7	17	19	21	23	25	27	29	32	34
8	19	22	24	26	29	31	34	36	38
9	22	24	27	30	32	35	38	41	43
10	24	27	30	33	36	39	42	45	48



Dobutamine (250mg/20mL)

Max concentration : 5 mg/ml

Solution : NSS, D5W, D5NS, D5S/2

Dosing instruction : Infusion Rate :2.5-15 µg/kg/min (up to 20 µg/kg/min if on beta-blockers)

Titrate dose by 2.5-5 µg/kg/min q 5-15mins to achieve a MAP≥65mmHg.

Infusion chart : Dobutamine or Dopamine 500 mg/500mL (1 mg/mL)

Dose (µg/kg/min)	Body weight (Kg)										
	35	40	45	50	55	60	65	70	75	80	85
	Drip rate (µd/min or mL/hr)										
3	6	7	8	9	10	11	12	13	14	14	15
4	8	10	11	12	13	14	16	17	18	19	20
5	11	12	14	15	17	18	20	21	23	24	26
6	13	14	16	18	20	22	23	25	27	29	31
7	15	17	19	21	23	25	27	29	32	34	36
8	17	19	22	24	26	29	31	34	36	38	41
9	19	22	24	27	30	32	35	38	41	43	46
10	21	24	27	30	33	36	39	42	45	48	51
12	25	29	32	36	40	43	47	50	54	58	61
14	27	34	38	42	46	50	55	59	63	67	71
16	34	38	43	48	53	58	62	67	72	77	82
18	38	43	49	54	59	65	70	76	81	86	92
20	42	48	54	60	66	72	78	84	90	96	102
22	46	53	59	66	73	79	86	92	99	106	112
24	50	58	65	72	79	86	94	101	108	115	122
26	55	62	70	78	86	94	101	109	117	125	133
28	59	67	76	84	92	101	109	118	126	134	143
30	63	72	81	90	99	108	117	126	135	144	153

Dopamine (500 mg/10mL)

Max concentration : 3.2mg/mL

Solution : NSS, D5W, D10W, D5NS, D5NS/2

Dosing instruction: Infusion Rate : Cardiac dose 5-10 µg/kg/min

Vasopressor dose 10-20 µg/kg/min

Titrate dose by 2-5 µg/kg/min q 5-15mins to achieve a MAP ≥65 mmHg.

Infusion chart : Dobutamine or Dopamine 1000 mg/500mL (2 mg/mL)

Dose (µg/kg/min)	Body weight (Kg)										
	35	40	45	50	55	60	65	70	75	80	85
	Drip rate (µd/min or mL/hr)										
3	3	4	4	5	5	5	6	6	7	7	8
4	4	5	5	6	7	7	8	8	9	10	10
5	5	6	7	8	8	9	10	11	11	12	13
6	6	7	8	9	10	11	12	13	14	14	15
7	7	8	9	11	12	13	14	15	16	17	18
8	8	10	11	12	13	14	16	17	18	19	20
9	9	11	12	14	15	16	18	19	20	22	23
10	11	12	14	15	17	18	20	21	23	24	26
12	13	14	16	18	20	22	23	25	27	29	31
14	15	17	19	21	23	25	27	29	32	34	36
16	17	19	22	24	26	29	31	34	36	38	41
18	19	22	24	27	30	32	35	38	41	43	46
20	21	24	27	30	33	36	39	42	45	48	51
22	23	26	30	33	36	40	43	46	50	53	56
24	25	29	32	36	40	43	47	50	54	58	61
26	27	31	35	39	43	47	51	55	59	62	66
28	29	34	38	42	46	50	55	59	63	67	71
30	32	36	41	45	50	54	59	63	68	72	77



Epinephrine (Adrenalin® 1mg/ml)

Max concentration : Undiluted (1mg/ml)

Solution : NSS, D5W, D5S

Dosing instruction : ACLS : 1 mg IV push q 3-5 min

Vasopressor Infusion : 1-10 µg/min (0.01-3 µg/kg/min) (Vasopressor dose)

Titrate dose by 0.01 µg/kg/min q 15 mins to achieve a MAP of ≥ 65 mmHg

Infusion chart : Epinephrine 10 mg/100mL (100µg/mL)

Dose (µg/kg/min)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.05	1	1	2	2	2	2	2	2	2
0.1	2	3	3	3	4	4	4	5	5
0.2	5	5	6	7	7	8	8	9	10
0.3	7	8	9	10	11	12	13	14	14
0.4	10	11	12	13	14	16	17	18	19
0.5	12	14	15	17	18	20	21	23	24
0.7	17	19	21	23	25	27	29	32	34
0.9	22	24	27	30	32	35	38	41	43
1	24	27	30	33	36	39	42	45	48
1.5	36	41	45	50	54	59	63	68	72
2	48	54	60	66	72	78	84	90	96
2.5	60	68	75	83	90	98	105	113	120
3	72	81	90	99	108	117	126	135	144

Esmolol (Esbloc® 100mg/10mL)

Max concentration : Undiluted (10mg/ml)

Solution : NSS, NS/2, D5W, D5S, D5S/2, D5NS

Dosing instruction : Bolus : 0.5-1 mg/kg in 1 min

Infusion: 0.05-0.3 mg/kg/min

Titrate dose by 0.05 mg/kg/min q 4 mins to desired effect. Taper off slowly.

Infusion chart : Esmolol 1000mg/100mL (10mg/mL)

Dose (mg/kg/min)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.025	6	7	8	8	9	10	11	11	12
0.05	12	14	15	17	18	20	21	23	24
0.075	18	20	23	25	27	29	32	34	36
0.1	24	27	30	33	36	39	42	45	48
0.125	30	34	38	41	45	49	53	56	60
0.15	36	41	45	50	54	59	63	68	72
0.175	42	47	53	58	63	68	74	79	84
0.2	48	54	60	66	72	78	84	90	96
0.225	54	61	68	74	81	88	95	101	108
0.25	60	68	75	83	90	98	105	113	120
0.275	66	74	83	91	99	107	116	124	132
0.3	72	81	90	99	108	117	126	135	144

**Fentanyl (100µg/2mL)****Max concentration** : Undiluted (50µg/mL)**Solution** : NSS, D5W**Dosing instruction** : Bolus: 0.35-0.5 µg/kg

Infusion rate: 0.5-10 µg/kg/h (25– 500 µg/h)

Titrate dose by 25 µg/h q 30 mins to achieve desired effect

If used with sedatives titrate to a RASS score of 0 to -2.

Infusion chart: Fentanyl 200 µg/100ml (2 µg/ml)

Dose (µg/kg/h)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.5	10	11	13	14	15	16	18	19	20
1	20	23	25	28	30	33	35	38	40
2	40	45	50	55	60	65	70	75	80
3	60	68	75	83	90	98	105	113	120
4	80	90	100	110	120	130	140	150	160
5	100	113	125	138	150	163	175	188	200

Infusion chart: Fentanyl 100 µg/100ml (1 µg/ml)

Dose (µg/kg/h)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.5	20	23	25	28	30	33	35	38	40
1	40	45	50	55	60	65	70	75	80
2	80	90	100	110	120	130	140	150	160
3	120	135	150	165	180	195	210	225	240
4	160	180	200	220	240	260	280	300	320
5	200	225	250	275	300	325	350	375	400

Heparin (25,000u/5ml)**Max concentration** : (25,000u/5ml)**Solution** : NSS (stable 24 hr), D5W (stable 8 hr)**Dosing instruction**:

Indication	IV Bolus dose (units/kg)	Max (units)	IV infusion rate (units/kg/hr)	Max (units/hr)
STEMI (Adjunct to fibrinolysis), NSTEMI, Unstable angina	60	4,000	12	1,000
AF (Cardioversion), DVT/PE, Arterial occlusion, Ergotism	80	10,000	18	1,800

Anticoagulation IV : 10,000 units Bolus, then 50 to 70 units/kg (5,000 to 10,000 units) q 4-6 hr**DVT/PE** : SC 333 units/kg then 250 units/kg q12 hr**Thromboprophylaxis** : SC 5,000 units q 8-12 hr**Antidote** : Protamine**Infusion chart : Heparin 25,000U (5 mL)/250 mL (100 units/mL)****Heparin 10,000U (2 mL)/100 mL (100 units/mL)**

aPTT	aPTT ratio	Action		Next aPTT
		Bolus	infusion rate	
<35 sec	น้อยกว่า 1.2 เท่า	80 units/kg	เพิ่ม 4 units/kg/hour	6 ชั่วโมง
35 to 45 sec	1.2-1.5 เท่า	40 units/kg	เพิ่ม 2 units/kg/hour	6 ชั่วโมง
46 to 70 sec	1.5-2.5 เท่า	ไม่ต้องปรับขนาด (therapeutic range)		6 ชั่วโมง หากเข้าเป้าหมายอีก ให้ดูครั้งต่อไปทุก 24 ชั่วโมง
71 to 90 sec	2.5-3.0 เท่า	-	ลด 2 units/kg/hour	6 ชั่วโมง
>90 sec	มากกว่า 3.0 เท่า	Hold 1 hr	ลด 3 units/kg/hour	6 ชั่วโมง



Lidocaine (40mg/2mL)

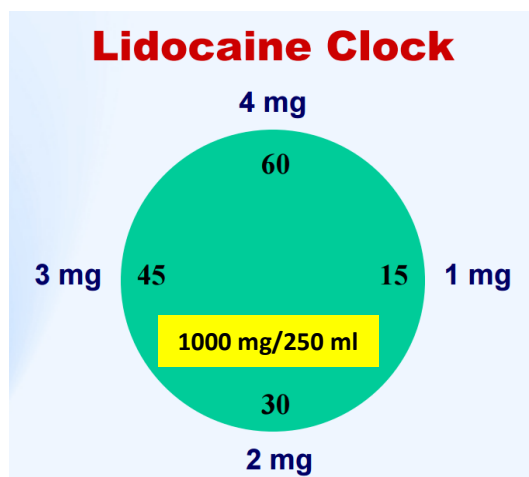
Max concentration : 20mg/mL

Solution : NSS, D5W

Dosing instruction : Bolus : 1-1.5 mg/kg over 2-3 mins (average 50-100mg)
; repeat dose 0.5-0.75 mg/kg in 3-5 min up to 3 mg/kg total
Infusion : 1-4 mg/min until arrhythmia terminated: stop infusion once rhythm stabilizes

Infusion chart: Lidocaine hydrochloride

Dose	Lidocaine 2000 mg + NSS or D5W up to 250 ml	Lidocaine 1000 mg + NSS or D5W up to 250 ml
1 mg/min	7.5 mL/hr	15 mL/hr
2 mg/min	15 mL/hr	30 mL/hr
3 mg/min	22.5 mL/hr	45 mL/hr
4 mg/min	30 mL/hr	60 mL/hr



Midazolam (Dormicum® 5mg/mL)

Max concentration : Undiluted(5mg/mL)

Solution : NSS, D5W

Dosing instruction : Bolus: 0.03-0.3 mg/kg (0.5-4mg) Infusion: 0.02-0.2 mg/kg/h (1-30 mg/h)
Titrate dose by 1 mg/h every 1 hour to achieve desired effect

Infusion chart : Midazolam 20mg/100mL (0.2mg/mL)

Dose (mg/kg/hr)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.02	4	5	5	6	6	7	7	8	8
0.04	8	9	10	11	12	13	14	15	16
0.06	12	14	15	17	18	20	21	23	24
0.08	16	18	20	22	24	26	28	30	32
0.1	20	23	25	28	30	33	35	38	40
0.125	25	28	31	34	38	41	44	47	50
0.15	30	34	38	41	45	49	53	56	60
0.2	40	45	50	55	60	65	70	75	80

Infusion chart : Midazolam 50mg/100mL (0.5mg/mL)

Dose (mg/kg/hr)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.02	2	2	2	2	2	3	3	3	3
0.04	3	4	4	4	5	5	6	6	6
0.06	5	5	6	7	7	8	8	9	10
0.08	6	7	8	9	10	10	11	12	13
0.1	8	9	10	11	12	13	14	15	16
0.125	10	11	13	14	15	16	18	19	20
0.15	12	14	15	17	18	20	21	23	24
0.2	16	18	20	22	24	26	28	30	32



Milrinone (Primacor® 10mg/10ml)

Max concentration : 0.2mg/mL

Solution : NSS, D5W, NS/2

Dosing instruction : Bolus: 50 µg/kg over 10 min

Infusion: 0.375 – 1 µg/kg/min (1.13 mg/kg/day)

Dose increases per CO/CI or SVO₂ and can be discontinued without weaning

Infusion chart : Milrinone 20mg/100mL (0.2 mg/mL)

Dose (µg/kg/min)	Body weight (Kg)										
	40	45	50	55	60	65	70	75	80	85	90
	Drip rate (µd/min or mL/hr)										
0.375	5	5	6	6	7	7	8	8	9	10	10
0.5	6	7	8	8	9	10	11	11	12	13	14
0.625	8	8	9	10	11	12	13	14	15	16	17
0.75	9	10	11	12	14	15	16	17	18	19	20
1	12	14	15	17	18	20	21	23	24	26	27

Morphine (10mg/ml)

Max concentration : 5mg/mL

Solution : NSS, D5W, D10W

Dosing instruction : Bolus: 2-5 mg (0.05-0.2mg/kg)

Infusion: 0.8-80 mg/h (0.01-0.03 mg/kg/hr)

Titrate dose by 1 mg/h q 30 min to achieve desired effect

Infusion chart : Morphine 20mg/100mL (0.2mg/mL)

Dose (mg/kg/h)	Body weight (Kg)								
	40	45	50	55	60	65	70	75	80
	Drip rate (µd/min or mL/hr)								
0.01	2	2	3	3	3	3	4	4	4
0.02	4	5	5	6	6	7	7	8	8
0.03	6	7	8	8	9	10	11	11	12

Nicardipine (2mg/2ml)

Max concentration : 0.2mg/mL

Solution : NSS, D5W, D5NS

Dosing instruction : Infusion: 5 - 15 mg/h

Titrate dose by 2.5 mg/h q 15 mins to achieve BP. Reduce infusion to 3-5 mg/hr after BP is controlled.

Infusion chart :

Dose(mg/h)	Nicardipine 10 mg/100mL	Nicardipine 20mg/100mL
	Conc 0.1 mg/ml	Conc 0.2 mg/ml
	Drip rate (µd/min or mL/hr)	
2.5	25	12.5
3	30	15
4	40	20
5	50	25
6	60	30
7	70	35
8	80	40
9	90	45
10	100	50
11	110	55
12	120	60
13	130	65
14	140	70
15	150	75



Nitroglycerine (50mg/10ml)

Max concentration : 0.4mg/mL

Solution : NSS, D5W

Dosing instruction : Infusion: 5 - 400 µg/min (0.25-1 µg/kg/min)

Titrate dose by 5 µg/min q 5 mins to achieve desired effect.

Infusion chart : Nitroglycerine 50mg/250mL (0.2 mg/mL)

Dose (µg /min)	Rate (mL/hr)		Dose (µg /min)	Rate (mL/hr)		Dose (µg /min)	Rate (mL/hr)
5	1.5		65	19.5		125	37.5
10	3		70	21		130	39
15	4.5		75	22.5		135	40.5
20	6		80	24		140	42
25	7.5		85	25.5		145	43.5
30	9		90	27		150	45
35	10.5		95	28.5		155	46.5
40	12		100	30		160	48
45	13.5		105	31.5		170	51
50	15		110	33		180	54
55	16.5		115	34.5		190	57
60	18		120	36		200	60

Nitroprusside (50mg/vial)

Max concentration : 1 mg/mL

Solution : D5W only

Dosing Instruction : Infusion: 0.5 – 10 µg/kg/min

Titrate dose by 0.25 µg/kg/min q 3 mins to achieve BP.

Infusion chart : Nitroprusside concentration 50mg/250mL (0.2mg/mL)

Dose (µg/kg/min)	Body weight (Kg)										
	40	45	50	55	60	65	70	75	80	85	90
	Drip rate (µd/min or mL/hr)										
0.5	6	7	8	8	9	10	11	11	12	13	14
0.75	9	10	11	12	14	15	16	17	18	19	20
1	12	14	15	17	18	20	21	23	24	26	27
1.25	15	17	19	21	23	24	26	28	30	32	34
1.5	18	20	23	25	27	29	32	34	36	38	41
1.75	21	24	26	29	32	34	37	39	42	45	47
2	24	27	30	33	36	39	42	45	48	51	54
2.5	30	34	38	41	45	49	53	56	60	64	68
3	36	41	45	50	54	59	63	68	72	77	81
3.5	42	47	53	58	63	68	74	79	84	89	95
4	48	54	60	66	72	78	84	90	96	102	108
4.5	54	61	68	74	81	88	95	101	108	115	122
5	60	68	75	83	90	98	105	113	120	128	135



Norepinephrine (Levophed® 4mg/4mL)

Max concentration : 1.28mg/mL

Solution : D5W only

Dosing Instruction : Infusion : 0.02-3 µg/kg/min or 0.5-30 µg/min

Titrate dose by 0.05 µg/kg/min q 3 mins to achieve a MAP of ≥65 mmHg.

Infusion chart : Norepinephrine standard dilution 4 mg in D5W 100mL (40 mcg/ml)																	
Rate (mL/hr)	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51
Dose (mcg/min)	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34
	Low dose			Recommended dose												High dose	

Infusion chart : Norepinephrine standard dilution 8 mg in D5W 100mL (80 mcg/ml)																	
Rate (mL/hr)	2	3	5	6	8	9	11	12	14	15	17	18	20	21	23	24	26
Dose (mcg/min)	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34
	Low dose			Recommended dose												High dose	

Dose (µg/kg/min)	Norepinephrine standard dilution 4 mg in D5W 100mL (40 mcg/mL)											
	Body weight (Kg)											
	40	45	50	55	60	65	70	75	80	85	90	
	Drip rate (µd/min or mL/hr)											
0.05	3	3	4	4	5	5	5	6	6	6	7	
0.1	6	7	8	8	9	10	11	11	12	13	14	
0.2	12	14	15	17	18	20	21	23	24	26	27	
0.3	9	10	11	12	14	15	16	17	18	19	20	
0.4	12	14	15	17	18	20	21	23	24	26	27	
0.5	15	17	19	21	23	24	26	28	30	32	34	
0.6	18	20	23	25	27	29	32	34	36	38	41	
0.7	21	24	26	29	32	34	37	39	42	45	47	
0.8	24	27	30	33	36	39	42	45	48	51	54	
0.9	27	30	34	37	41	44	47	51	54	57	61	
1	60	68	75	83	90	98	105	113	120	128	135	

Propofol (Diprivan® 200mg/20mL)

Max concentration : Undiluted (10mg/mL) **Min concentration :** 2mg/mL

Solution : D5W

Dosing Instruction : Bolus : 1-3mg/kg

Infusion : 5-100 µg/kg/min

Titrate dose by 5 µg/kg/min q 5 mins to achieve RAAS score 0 to -2 if use with paralytics, titrate to a bispectral index (BIS) of 40 to 60

Infusion chart : Propofol 1000mg/100mL (10mg/mL)

Dose (µg/kg/min)	Body weight (Kg)											
	40	45	50	55	60	65	70	75	80	85	90	
	Drip rate (µd/min or mL/hr)											
5	1	1	2	2	2	2	2	2	2	3	3	
10	2	3	3	3	4	4	4	5	5	5	5	
15	4	4	5	5	5	6	6	7	7	8	8	
20	5	5	6	7	7	8	8	9	10	10	11	
25	6	7	8	8	9	10	11	11	12	13	14	
30	7	8	9	10	11	12	13	14	14	15	16	
35	8	9	11	12	13	14	15	16	17	18	19	
40	10	11	12	13	14	16	17	18	19	20	22	
45	11	12	14	15	16	18	19	20	22	23	24	
50	12	14	15	17	18	20	21	23	24	26	27	
55	13	15	17	18	20	21	23	25	26	28	30	
60	14	16	18	20	22	23	25	27	29	31	32	
65	16	18	20	21	23	25	27	29	31	33	35	
70	17	19	21	23	25	27	29	32	34	36	38	
75	18	20	23	25	27	29	32	34	36	38	41	



Rocuronium (Esmeron® 50mg/5ml)

Max concentration : Undiluted (10mg/ml)

Solution : NSS, D5W, D5S

Dosing Instruction : Bolus: 0.6-1.2mg/kg

Infusion: 0.4-0.7 mg/kg/h

Titrate dose to achieve desired effect

Infusion chart : Rocuronium 100mg/100mL (1mg/ml)

Dose (mg/kg/h)	Body weight (Kg)										
	40	45	50	55	60	65	70	75	80	85	90
	Drip rate (µd/min or mL/hr)										
0.4	16	18	20	22	24	26	28	30	32	34	36
0.45	18	20	23	25	27	29	32	34	36	38	41
0.5	20	23	25	28	30	33	35	38	40	43	45
0.55	22	25	28	30	33	36	39	41	44	47	50
0.6	24	27	30	33	36	39	42	45	48	51	54
0.65	26	29	33	36	39	42	46	49	52	55	59
0.7	28	32	35	39	42	46	49	53	56	60	63

Vecuronium (Vecuron® 10 mg/vial)

Max concentration : 2mg/ml

Solution : NSS, D5W, D5NS

Dosing Instruction : Bolus : 0.08-0.1 mg/kg

Initial : 0.8-2 µg/kg/min

Titrate dose by 0.2 µg/kg/min q 10 mins to achieve desired effect.

Infusion chart : Vecuronium 100mg/250mL (0.4mg/ml)

Dose (µg/kg/min)	Body weight (Kg)										
	40	45	50	55	60	65	70	75	80	85	90
	Drip rate (µd/min or mL/hr)										
0.8	5	5	6	7	7	8	8	9	10	10	11
1	6	7	8	8	9	10	11	11	12	13	14
1.2	7	8	9	10	11	12	13	14	14	15	16
1.4	8	10	11	12	13	14	15	16	17	18	19
1.6	10	11	12	13	14	16	17	18	19	20	22
1.8	11	12	14	15	16	18	19	20	22	23	24
2	12	14	15	17	18	20	21	23	24	26	27



	Acyclovir sodium	Amikacin sulfate	Amiodarone HCl	Ampicillin/Sulbactam	Azithromycin	Calcium Gluconate	Cefazolin Sodium	Cefepime HCl	Cefotaxime Sodium	Ceftazidime Sodium	Ceftriaxone sodium	Ciprofloxacin	Clindamycin	Co-trimoxazole	Dobutamine HCl	Dopamine HCl	Epinephrine HCl	Esmolol HCl	Fentanyl citrate	Fluconazole	Furosemide	Gentamicin sulfate	Heparin Sodium	Hydrocortisone	Imipenem/cilastatin	Insulin Regular	Labetolol	Levofloxacin	Lidocaine HCl	Magnesium Sulfate	Meropenem	Metronidazole HCl	Midazolam HCl	Milrinone Lactate	Morphine sulphate	Nicardipine HCl	Nitroglycerine	Nitroprusside Sodium	Norepinephrine	Octreotide Acetate	Phenytoin Sodium	Potassium Chloride	Propofol	Sodium Bicarbonate	Vancomycin HCl	Vecuronium Br			
Amiodarone HCl	I	C		I	I	I	I	C	I	I	I	C	C	I	C	C	C	C	C	C	I	C	I	I	I	C	C	I	C	I	I	C	C	C	C	C	C	I	C	C	I	C		I	C	C			
Atracurium	C	C	C	C	C	C	C		C	I	C		C	C	C	C	C	C	C	C	I	C	I	C	C	C	C	C	C	C	C		C	C	C	C	C	C	I	C	C	I	C	I	C				
Cisatracurium	C	C	C	C	C	C				C	C	C	C	I	C	C	C	C	I	C	C	C	C	C	C		C	C	C	C			C	C		C	C	I	C			C	I	I	C				
Dobutamine HCl	I	C	C	I	C	C	I	I	I	I	I	C	C	I		C	C	C	C	C	I	C	I	I	I	I	C	C	C	C	C		C	I	C	C	C		C	C	I	C	C	I	C	C			
Dopamine HCl	I	C	C	C	C	C	I	I	C	C	C	C	C	I	C		C	C	C	C	I	C	C	C	C	I	C	C	C	C		C	C	C	C	C	C	C	C	I	C	C	I	C	C				
Epinephrine HCl	I	C	C	C	C	C			C	C	C		C	I	C	C		C	C		C	C	C	C	C	I	C	C	C	C		C	C	C	C	C	C	C	C	C	I	C	C	I	C	C			
Esmolol HCl	I	C	C	C	C	C	C	C	C	C	C		C	I	C	C	C		C	C	I	C	C	C	C	C	C	C	C	C	C	C	C	I	C	C	C	C	C	C	I	C	C	C	C	C			
Fentanyl citrate	C	C	C	C	I	C	C		C	C	C		C	I	C	C	C	C		C	C	C	C	C	C	C	C	C	C	C	C		C	C	C	C	C	C	C	C	C	I	C	C	C	C	C		
Heparin Sodium	C	I	I	C	C	C	C		C	C	C	I	C	I	I	C	C	C	C	C	C	I		C	C	C	I	I	C	C	C	C	C	C	C	C	I	C	C	C	C	I	C	C	C	I	C	C	
Insulin Regular	C	C	C	C		C	C	C	C	C	C		C	I	C	I	I	C	C	C	I	I	C	C	C		I	C	C	C	C	C	C	C	I		C	C	I	C	I	C	C	C	C	C	C		
Labetolol	I	C	C	C	C	C	I		I	C	I	C	I	I	C	C	C	C	C	C	I	C	I	I	C	I		C	C	C	C	C	C	C	C	I		C	C	C	C	I	C	C	C	C	C		
Lidocaine HCl	I	C	C	C	C	C	C		C	C	C	C	C	I	C	C	C	C	C	C	C	C	C	C	C	C	C	C		C		C	C	C	I	C	C	C	C	C	I	C	I	C	C	C	C		
Magnesium Sulfate	C	C	C	C	C	C	C	I	C	C	I	I	C	I	C	C	C	C	C	C	I	C	C	I	C	C	C	C	C	C		C		C	C	C	C	C	C	C	C	I	C	C	C	C	C		
Midazolam HCl	I	C	C	I	I	C	C	I	C	I	C	C	I	I	I	C	C	C	C	C	I	C	C	I	I	C	C	C	C	C	C		C		C		C	C	C	C	C	I	C	I	I	C	C	C	
Milrinone Lactate	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	I	C	C	I	C	C	C	I	C	C	C	I	C	C	C	C		C	C	C	C	C	C	C	I	C	C	C	C	C		
Morphine sulphate	I	C	C	C	I	C	C	I	C	C	C		C	I	C	C	C	C	C	C	I	C	C	C	C	C	C	C	C	C	C	C	C	C		C	C	C	C	C	C	I	C	C	C	C	C	C	
Nicardipine HCl	I	C	C	I	I	C	C	C	C	C	C	C	C	I	C	C	C	C	C	C	I	C	I	I	I		C	C	C	C	C	I	C	C	C		C	C	C	C	I	C	I	I	C	C	C	C	
Nitroglycerine	C	C	C	C	C	C	C		C	C	C		C	I	C	C	C	C	C	C	I	C	C	C	C	C	C	C	I	C	C		C		C	C	C		C	C	C	I	C	C	C	C	C	C	
Nitroprusside Sodium	I	C	I	C	C	C	C		C	I	C		C	I	C	C	C	C	C	C	C	C	C	C	I	C	C	I	C	C	C		C		C	C	C		C	C	C	I	C	C	C	C	C	C	
Norepinephrine		C	C	C	C	C	C		C	C	C		C	I	C	C	C	C	C	C	I	C	C	C	C	I	C		C	C	C	C	C	C	C	C	C		C	C		C	I	C	C	I	C	C	C
Pancuronium HBr	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	I	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	I	C	C	C	C				
Propofol	C	I				C	C	I	C		C	I	C	I	C	C	I	C	C	C	C	I	C	C	C	C	C	I	I	C		I	C	C	C		C	C	C		I	C		C	I	I			
Vecuronium Br	I	C	C	C	C	C	C	I	I	C	C	C	C	C	C	C	C	C	C	C	I	C	C	C	I	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	I	C	I	C	C				

IV Compatibility



Compatible



Incompatible



No data